Optimal Control for Improving Assistive Exoskeletons

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***Abstract*—Falls during daily ambulation are a leading cause of injury in diverse clinical populations, motivating advances in exoskeleton control, balance assistance, and personalized data-driven adaptation. In this work, we present two complementary studies. First, we apply a Linear Quadratic Regulator (LQR) to a MuJoCo humanoid model standing on a moving platform, simulating scenarios such as riding a moving vehicle (e.g. a bus). Through simulation experiments under a range of perturbation conditions, including setups with a support railing and a simplified hip exoskeleton, we demonstrate that the LQR controller consistently maintains balance and stability. Second, we leverage iterative Linear Quadratic Gaussian (iLQG) trajectory optimization seeded with individual motion-capture kinematics to, in the future, generate synthetic data by modelling a person's gait pattern in simulation. This would enable the future placement of virtual Inertial Measurement Units (IMUs) onto that walking model, which is useful in situations where such equipment is not available. These results demonstrate that both LQR balancing and iLQG tracking are promising approaches for making exoskeleton control easier and safer.**

I. Introduction

Exoskeletons hold significant promise for individuals with lower limb impairments, enabling greater independence in mobility and supporting the performance of daily ambulation tasks without reliance on wheelchairs or external assistance. A key focus of exoskeleton research is enhancing user safety by providing fall protection. Wearable lower-limb exoskeletons can intervene faster than human reflexes by applying corrective torques at the ankle, knee, and hip joints, significantly improving standing balance, reducing user effort and preventing falls [2]. Yet two critical challenges remain: how to design real-time controllers that maintain stability under unpredictable disturbances, and how to obtain sufficient individualized gait data to personalize those controllers.

In this work, we address these challenges through two distinct studies. First, we develop a Linear Quadratic Regulator (LQR) for humanoid balance: by linearizing the MuJoCo humanoid dynamics around the upright equilibrium and solving the continuous-time Riccati equation, we compute state-feedback gains that regulate the center of mass under random oscillations of a moving platform, simulating the environment of a moving vehicle (i.e. bus, train). We vary elements across scenarios, adding support railings and a basic exoskeleton structure to study how they help the humanoid balance.

Second, we tackle the problem of seeding iterative Linear Quadratic Gaussian (iLQG) trajectory optimization with motion-capture kinematics. While much work considers sim-to-real transfers, there is also value in performing the inverse: real-to-sim. Mapping real kinematic data of a person walking onto a simulation model that interacts with the world through a physics engine could enable the placement of virtual IMUs onto that walking model. With kinematics extraction being made possible with a simple cellphone video recording [13], these virtual IMUs would further lower the requirement of gathering personalized data in a lab environment.

II. Related Work

A. Optimal Feedback Control for Balance Assistance

Maintaining balance is a fundamental challenge in many robotics applications, such as humanoid robotics and wearable exoskeletons. Early work in humanoid control introduced optimal feedback strategies such as the Linear Quadratic Regulator (LQR) to stabilize simplified models (e.g. inverted pendulums), enabling robots to recover from pushes via appropriate ankle torques or stepping responses [7]. The concept of the capture point was proposed as an optimal step location to arrest a fall [7]. Building on these foundations, model predictive control (MPC) approaches were developed to handle balance in real time.

Most controllers simplify to an inverted pendulum, limiting their ability to be used for real-time MPC on exoskeleton hardware. Data-driven methods are beginning to complement model-based designs: Molinaro et al. introduced a task-agnostic controller that estimates biological joint moments in real time, generalizing assistance across diverse activities without predefined gait modes [3]. These ML-driven methods hint at future exoskeletons that maintain balance in a wide range of motions.

B. Motion-Capture-Informed Synthetic Data Generation

Another research direction is generating synthetic, personalized gait data for controller design and tuning. Traditional exoskeleton control relies on lab-recorded gait patterns or predefined trajectories, but emerging methods use trajectory optimization and learning to synthesize new motions tailored to the individual.

Todorov and Li's iterative Linear Quadratic Gaussian (iLQG) algorithm extends LQR to nonlinear systems [6], like one would encounter with a human or humanoid. Deep reinforcement learning (RL) has also been applied for gait synthesis. DeepMimic (Peng et al., 2018) used reference motion clips to train policies that imitate human locomotion and acrobatics in simulation [10].

Synthetic gait data enables pre-deployment testing of exoskeleton algorithms in simulation and can inform controller personalization with minimal human trials. Model-based simulations predict how gait adapts under varied conditions, identifying potential balance issues and optimal assistance strategies [5]. However, ensuring fidelity of synthetic data remains a challenge, with poor model accuracy leading to a sim-to-real gap. GP-iLQG methods mitigate this by continually correcting the model with limited real data [9], but still require some human-in-the-loop calibration. The key challenge is reducing the motion capture and computation required while ensuring that synthetic gaits translate safely and effectively to real exoskeleton hardware.

III. Problem Formulation

A. Humanoid Model

We employ the MuJoCo implementation of the DeepMind Control Suite's 3D biped "Humanoid," which represents an adult-sized figure with anatomically realistic mass and inertia. The model features:

- **Degrees of Freedom:** 27 joints, including a 6 DOF floating pelvis, 3 DOF torso (spine), and limb: 3 DOF hips, 1 DOF knees, 2 DOF ankles, 2 DOF shoulders, and 1 DOF elbows.
- **Actuation:** 21 torque-controlled hinge motors (all non-floating joints) with default gains and damping tuned for human-like responsiveness.

The simulator enforces rigid, no-slip foot contacts on a moving support surface and provides full state feedback (joint angles, velocities, base pose) alongside measured platform motion.

B. Control Objective

Our primary goal is to synthesize a full-body LQR controller in MuJoCo that maintains upright balance on a dynamic, vehicle-like floor. The other goal is to employ the iterative Linear Quadratic Gaussian (iLQG) algorithm to reproduce an individual's gait patterns in simulation. While this study targets all 21 joints to establish baseline balance performance, our longer-term vision is to distill the approach to a lower-limb-only controller—directly applicable to assistive exoskeletons.

C. Discrete LQR Overview

Consider the discrete-time linear system

$$x_{k+1} = A\,x_k + B\,u_k, \quad x_k \in \mathbb{R}^n,\ u_k \in \mathbb{R}^m. \tag{1}$$

Choose symmetric positive–definite weighting matrices

$$Q \succ 0, \quad R \succ 0,$$

and define the one-step cost

$$J(x_k, u_k) = x_k^\top Q\,x_k + u_k^\top R\,u_k. \tag{2}$$

For any state-feedback policy $\pi\colon \mathbb{R}^n \to \mathbb{R}^m$, the associated cost-to-go is

$$V^\pi(x_0) = \sum_{k=0}^{\infty} J\big(x_k, \pi(x_k)\big) \tag{3}$$

Kalman's result states the optimal policy is

$$\pi^*(x) = -K\,x,$$

with value function

$$V^*(x) = x^\top P\,x, \tag{4}$$

where P Solve the Discrete-time Algebraic Riccati Equation

$$P = A^\top P\,A + Q - A^\top P\,B\,(R + B^\top P\,B)^{-1} B^\top P\,A, \tag{5}$$

and the gain

$$K = (R + B^\top P\,B)^{-1}\,B^\top P\,A. \tag{6}$$

Thus find the control input

$$u_k = -K\,x_k \tag{7}$$

D. ILQG Overview

The iLQG algorithm iterates through five main stages, as seen in Fig. 1. First, we initialize the controller with a nominal trajectory. For this nominal trajectory, cold-starting solely using the real trajectory caused significant issues that could not be resolved with further shaping of the costs. Therefore, the output of an existing MPC walking strategy from [14] was used as both a warm-starting trajectory and to improve the overall tracking of real data in the simulation environment.

Second, after initializing with the MPC nominal trajectory, we linearize the dynamics and compute the cost derivatives for the current trajectory. The Jacobians are provided directly by MuJoCo, which are then also used to calculate the cost derivatives. Third, using these derivatives, we perform a backward pass (Riccati recursion), essentially solving a dynamic programming problem to get a sequence of feedback gains to minimize the cost-to-go locally.

Fourth, a line search is performed for a specific candidate trajectory (and control input sequence), and if the total cost is less than the nominal cost $J_{nominal}$, this trajectory is accepted as the new nominal. Once a new nominal trajectory is found, this control sequence is immediately rolled out, and the process begins anew. If a candidate trajectory is greater than $J_{nominal}$, then the step size is decreased and a new candidate trajectory is generated, until a trajectory with a lower cost than $J_{nominal}$ is found.

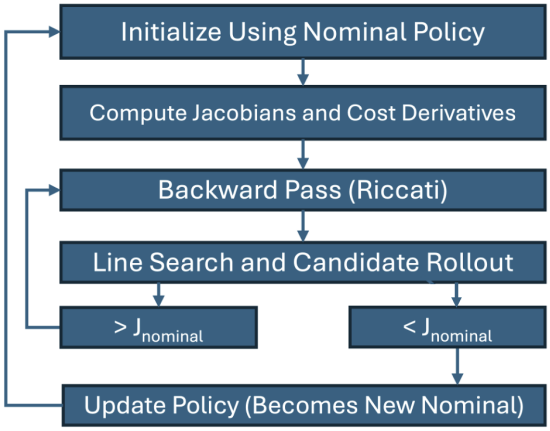

Fig. 1. iLQG Algorithm

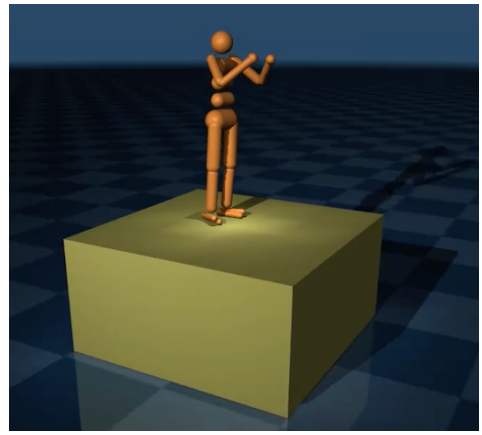

Fig. 2. Experiment 2: Setup

IV. Simulation Setup and Experiments

A. LQR Experiments

*1) **Experiment 1: Balancing with foot position constraints**:* To assess the feasibility of using a Linear Quadratic Regulator (LQR) for balance control, we designed a simplified experimental setup. The simulation involved a MuJoCo humanoid model III-A), placed on a moving platform. To simulate external perturbations, random velocities were applied in the Y-direction of the platform, mimicking forward and backward disturbances.

The LQR cost function was formulated with the following objectives:

1) Maintaining balance – penalizing states that lead to instability or falling,
2) Minimizing deviations from an upright stance – especially in the lower body joints,
3) Regulating upper body motion – ensuring that arm and torso movements remain within reasonable bounds to aid balance without excessive motion.

Equality constraints were added in the XML file to attach the humanoid's feet to the moving platform. This setup allowed us to evaluate the controller's ability to stabilize the humanoid under disturbances. We referred to the methodology outlined in [16] but adapted it to suit our specific objectives. All setup MuJoCo XML models were written referring to MuJoCo XML documentation [17].

*2) **Experiment 2: Balancing with free foot positions**:* Building on experiment 1, we removed the equality constraints between the feet and the moving platform to teach the humanoid to balance using varying foot placement. The rest of the setup remained the same and is pictured in Figure 2. We tested forward-backward perturbations to study the robustness of the LQR controller.

*3) **Experiment 3: Balancing with the aid of a railing**:* To simulate real-life scenarios where humans balance in dynamic environments, we introduced a railing attached to the moving platform. This addition was intended to encourage the humanoid to use the railing as a support mechanism during perturbations.

The railing was designed to be approximately 2/3 times height of the humanoid, providing a reachable aid for balance from its starting position, where the humanoid was initialised with hands held up to its chest. This setup is pictured in Figure 3. To reflect this setup in the controller design, the LQR cost function was augmented with an additional term aimed at penalizing deviations of both hands from the surface of the railing. Initially, this term penalized the vertical (Z-axis) distance between the humanoid's hands and the railing, encouraging the controller to position the hands close or on the support.

To improve both the effectiveness and accuracy of this cost term, it was later refined to use the Euclidean distance between the center of each hand and a point anywhere on top of the railing surface. This more precise measure allowed the controller to better exploit contact forces at the hands to aid in overall stabilization.

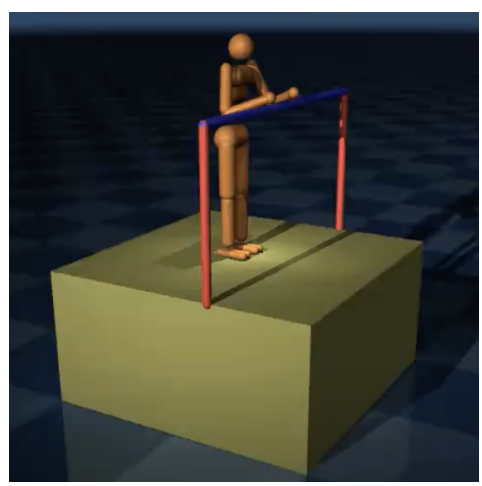

Fig. 3. Experiment 3: Setup

*4) **Experiment 4: Balancing with an exoskeleton**:* As a final experiment, we designed a hip exoskeleton to provide external support to the humanoid during balance control. Modeling a real exoskeleton presented various challenges, but the goal was to design a skeleton that could, to some extent, emulate exoskeleton behavior. Through this, we aimed to test the feasibility of our approach and explore its potential applications.

As illustrated in Figure 4, the exoskeleton comprised of a thin rod positioned above the humanoid's pelvis at the back, connected to two perpendicular rods extending along either thigh, joined by hinge joints with the same range of motion as the humanoids hips. The rods ran parallel to the thighs and terminated just above the knees, where they were joined to the lower thigh via small cylindrical connectors. We disabled the humanoid's default hip actuators, and added actuators on the exoskeleton's hinge joints.

To ensure coupling between the exoskeleton and the humanoid's legs, we imposed weld equality constraints between the thighs and connectors. This modification forced the humanoid to rely on the exoskeleton's actuation to generate hip movements. Contact forces were disabled between the exoskeleton and the lower waist of the humanoid as it led to large reactive forces which kept toppling the humanoid.

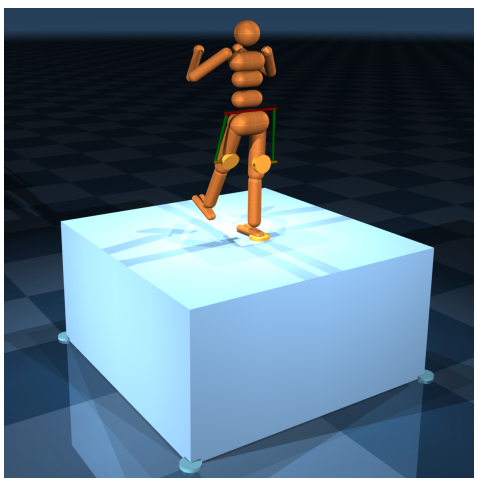

Fig. 4. Experiment 4: Setup. Contact forces are for visualisation purposes only.

B. ILQG Experiment

To get the nominal walking trajectory for warm-starting the iLQG tracking, the MuJoCo MPC walking policy was used. The cost function for this walking trajectory contains elements like the height, velocity, balance, and posture along with other factors like a measure for a walking gait and the general control cost. These cost function elements were iteratively increased and decreased until a satisfactory walking behavior was reached, after which the keypoints were recorded from a standing start, similar to how the tracking problem begins. The walking behavior roughly matched the same cadence, or footstep contact timing, as that of the tracking problem.

For the real data, a walking gait was selected from the CMU mocap dataset (mocap.cs.cmu.edu) [15], namely subject 137, trial 40, for which the keypoints had already been prepared in [14]. The goal was to create a cost function that would merge these two sources of motion, so a single stage cost function was created to penalize both deviation from the MPC-generated keypoints and the CMU mocap gait:

$$J = \sum_{t=0}^{T-1} \Big[\underbrace{w_1 \|\mathbf{p}_t - \mathbf{p}_t^{\text{mocap}}\|^2 + w_2 \|\mathbf{p}_t - \mathbf{p}_t^{\text{offline}}\|^2}_{\text{stage cost}} \underbrace{+ w_3 \|\mathbf{u}_t\|^2}_{\text{stage cost}} \Big] + \underbrace{w_4 \|\mathbf{x}_T - \mathbf{x}_{\text{target}}\|^2}_{\text{terminal cost}} \tag{8}$$

Here, $p_t^{offline}$ are the MPC keypoints, p_t^{mocap} are the real keypoints, and p_t is are the simulated MuJoCo keypoints at time t. 16 keypoints were chosen throughout the humanoid model, with 3 keypoints in the torso, 6 in the legs, 6 in the arms, and 1 in the head. The keypoint for the head was removed from the tracking cost, leading to an improvement in performance. The control cost is the u_t term, and an additional terminal cost was included to ensure that the final body state would be stable. However, since only an x_{target} body state was known for the MPC walking gait, using these values had a negative effect on the overall tracking performance, and was removed as part of this experiment. In the future, creating an x_{target} using real kinematics may be helpful.

To match the MPC and real trajectories in time, they were first trimmed to the same duration and number of frames. The start was also aligned to the standing position. The weights were initially set to $w_1 = 1$, $w_2 = 1$, and $w_3 = 0.1$. These were then manually tuned until the walking trajectory qualitatively resembled the original mocap movement for multiple seconds. A planning horizon of 0.3750 seconds and the MuJoCo "ImplicitFast" integrator achieved the best performance.

V. Results and Discussion

Due to the visual nature of this work, the results are best viewed as videos at the following location: https://tinyurl.com/OCRL-Project.

A. LQR

As illustrated in Fig. 5, Experiment 1 exhibited the greatest robustness to high-speed perturbations. This was attributed to the application of equality constraints at the feet, which enforced constant contact with the platform and enabled stable load distribution across both feet. Thus, the humanoid in Experiment 1 could withstand random perturbations in the range of -5.1 to 5.1 m/s at each timestep.

In Experiment 2, the removal of these constraints exposed the humanoid's center of mass (COM) to direct disturbances. Without the stabilizing effect of constrained foot placement, external perturbations were transmitted through the ankle's DOFs, often resulting in a loss of balance, mostly due to

a disconnect between the feet and platform. A large dip in maximum perturbation speed is observed from Experiment 1 to 2, where the highest speed at which the humanoid can balance goes from 5.1 to 0.36 at experiment 2, due to the removal of equality constraints.

In Experiment 3, the inclusion of a support railing slightly improved the humanoid's stability under perturbations. The humanoid leveraged contact forces and friction at the rail interface to react to disturbances, thus tolerating higher perturbation speeds than in the unsupported scenario. Experiment 3 sees a slight increase in the max perturbation speed (from 0.36 to 0.45). Higher contact forces at the railing also push away the humanoid, causing it to fall back and topple over. Further contact tuning is required for these bodies so that better support can be provided to the humanoid by the railing.

In Experiment 4, the addition of exoskeleton weight slightly altered the balancing problem. Given the inherent challenges of modeling an exoskeleton accurately in MuJoCo, we anticipated that the results from this task might not fully reflect realistic or physically accurate behavior. Nevertheless, the exoskeleton was able to balance under a designed cost function and constraint setup for perturbation velocities ranging from –0.1 to 0.1 m/s.

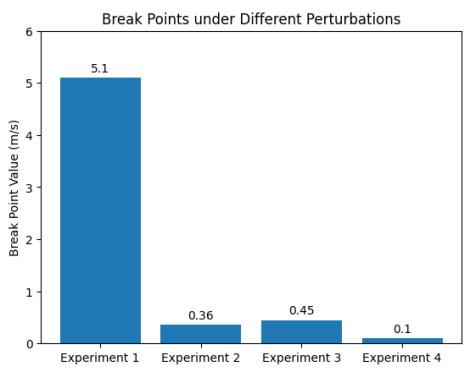

Fig. 5. Max Perturbation Velocity by Experiments

To verify whether our XML model correctly coupled the thigh motion to the exoskeleton actuator, we plotted hip joint angles and control inputs. As shown in Fig. 6, the two signals exhibited similar visual trends, suggesting that the actuator influenced thigh movement as intended. The high frequency in the graph can be attributed to small, frequent changes in the control inputs and angles to balance amidst random, high frequency perturbations.

We acknowledge that the exoskeleton implemented here is a highly simplified representation of a real-world system, lacking many of the physical interactions and complexities that exist in practice. While this mimics exoskeleton behavior to some extent, it does not constitute a true biomechanical or robotic exoskeleton model, and this limitation should be kept in mind. That said, the results remain encouraging and provide a useful foundation for future work in modeling exoskeletons and developing control strategies for humanoid balance.

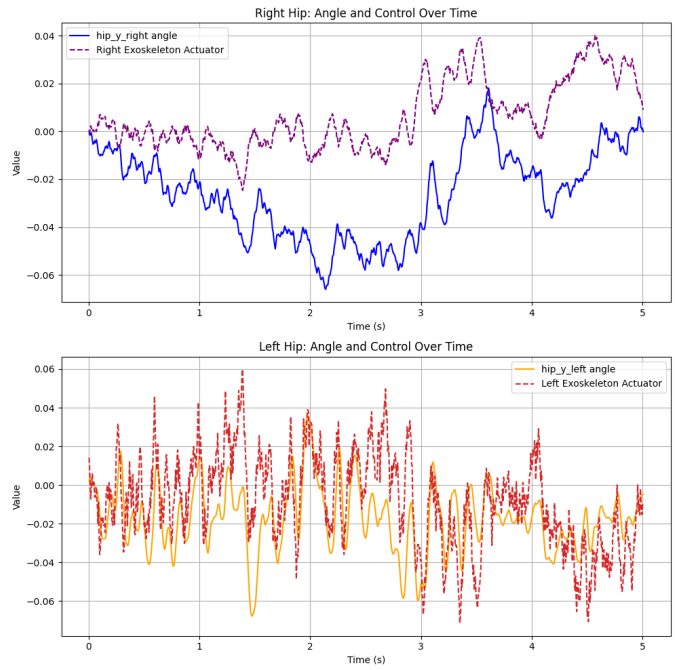

Fig. 6. Hip Angles and Exoskeleton Control across time

B. iLQG

After manually tuning the cost function weights until the iLQG controller was able to qualitatively track the provided CMU mocap data keypoints (subject 137, trial 40), 30 independent tracking trials were recorded to get a better understanding of the average performance. For each trial, the seconds were counted from simulation initialization until the simulation keypoints depicted a large, visual deviation from the real keypoints. For the 30 trials, the mean time until significant deviation was 4.8 seconds (± 1.06 SD) out of the seven total seconds of walking data, as seen in Fig. 7, with the red numbers representing every second in the simulation, and the white dots representing the p_t^{mocap} keypoints.

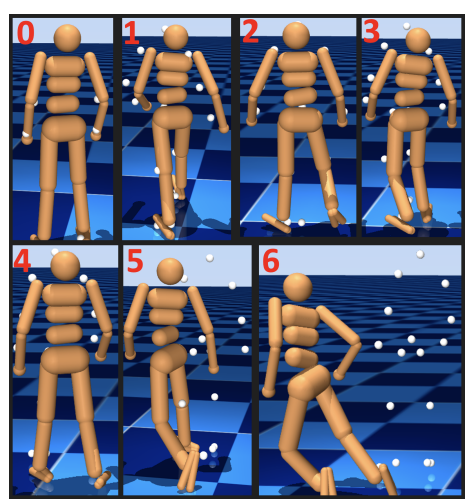

Fig. 7. iLQR Tracking Results

After about five seconds, the difference between the simulated keypoints p_t and $p_t^{offline}$ / p_t^{mocap} grew too large, leading to a large over-correction event which destabilized the humanoid model and caused it to fall. In the worst case, divergence occurred as early as two seconds, so more work needs to be done in adapting the cost function to make this method more robust.

VI. Future Work

A. LQR Balance

Our LQR controller has two main limitations. First, our current cost function only penalizes deviation from the upright pose. We can better balance performance by fine-tuning our cost function to be much more compreshensive, introducing additional terms such as weighted penalties on upper-body torques to limit excessive arm or torso motion or explicit energy penalties to encourage low-power, stable strategies. In addition, we could implement a more sophisticated penalty to control input, such as restricting the motions contributed by ankles and wrists.

Second, because we only actuate the hip–thigh joint with the simplified exoskeleton model, we overlook the balance contributions from the knee and ankle. In future work, we can implement a more sophisticated exoskeleton model with multijoint actuation so that torques can be optimally distributed throughout the lower limb to improve stability and performance. Additionally, our current model does not account for contact forces between the exoskeleton and the body, surface friction, or the impact of added exoskeleton weight, which could shift the humanoid's center of mass (COM) and affect balance. Future work must consider the impact of these to effectively simulate and study exoskeleton control for balancing and stability.

Future directions will also have to address how to transfer the learned LQR knowledge to the current state of the art exoskeleton controllers, which are based on Temporal Convolutional Networks (TCNs). Potentially the LQR simulation data can be used as a form of synthetic data to further train the TCN on balancing methods. Another avenue might be using LQR as a form of safety controller that gets initiated when the TCN fails to keep the human balanced, especially for unknown situations.

B. iLQG Tracking

One limitation of the current iLQG real-to-sim tracking is that only 16 keypoints were tracked, and future work should consider the full kinematic chain between real data and the simulation. There may also be value in foregoing keypoints altogether and instead using joint angles, since these are more independent of the simulation humanoid model, with human to simulation humanoid misalignment being a potential contributor to a drop in tracking accuracy.

The MPC-generated gait could also be "blended" with the real mocap data by using an interpolation method or smoothing spline, instead of incorporating them in the cost function independently. Also, since most of the failure cases resulted from a very strong control input at the end, adding more constraints to the maximum humanoid actuator torque could help prevent strong deviations.

The next steps would be to add virtual sensors (IMUs) to the simulation model, and then compare that data to a walking dataset where IMU data is available, to analyze the feasibility of the iLQG tracking approach in generating such synthetic data. If proven feasible, then this approach could also allow for the rapid testing of optimal sensor placement along the body.

VII. Conclusion

In this work, we examined two strategies with the potential to improve exoskeleton control: Using LQR for balancing and iLQG for real-to-sim trajectory tracking. In a MuJoCo simulation environment created to model a moving public-transport vehicle, we showed that LQR could be used to help balance a person subjected to a vehicle's perturbations while standing. The LQR controller was able to consistently maintain upright stability across a variety of disturbances and model conditions provided full-body actuation. We also introduced a preliminary exoskeleton model in MuJoCo to show the feasibility of this approach given more actuation constraints.

Additionally, the iLQG-based trajectory tracking method was successful at transferring a person's individual gait pattern to the simulation model for a mean of 70% of the total trajectory. With further improvements, like moving from keypoints to joint angles, this real-to-sim transfer has the potential to generate synthetic, personalized data within simulation, lessening the need for lab experiments. Future work will focus on transferring these techniques to physical exoskeleton hardware and controllers.

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